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**School of
Electronics and Communication Engineering**

**Minor Project Report on
Fusion of Localization Inputs**

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CERTIFICATE

This is to certify that project entitled “**Fusion of localization inputs**” is a bonafide work carried out by the student team of **Ismail Golsangi, Shraddha Soudagar, Trupti G C, Tanishka Rampure**. The project report has been approved as it satisfies the requirements with respect to the minor project work prescribed by the university curriculum for BE (VI semester) in School of Electronics and Communication Engineering of KLE technological University for the academic year 2024-2025.

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-The project team

ABSTRACT

Accurate and robust object detection with distance estimation is critical for safe autonomous driving, particularly in complex urban environments. This project proposes a sensor fusion-based approach that leverages both 2D vision (RGB images) and 3D spatial data (LiDAR point clouds) to detect objects and estimate their real-world distances. The system integrates the state-of-the-art YOLOv8 object detection model with LiDAR point cloud clustering to achieve high-precision object localization and distance computation. It utilizes the KITTI Vision Benchmark dataset, which provides synchronized RGB images, LiDAR scans, and calibration data for real-world autonomous driving scenarios. The pipeline begins with the YOLOv8m model performing object detection on RGB images to obtain 2D bounding boxes. These detections are then geometrically projected into 3D space using calibration matrices from the KITTI dataset. Simultaneously, the raw LiDAR binary point cloud is loaded, transformed into the camera coordinate frame, and projected onto the image plane for correspondence with the 2D detections. To segment objects in the point cloud, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is applied, isolating clusters that likely correspond to physical objects. Each YOLO-detected bounding box is matched with a LiDAR cluster by computing the IoU between the projected LiDAR cluster and the image. A high IoU score indicates strong spatial overlap, enabling accurate association of 2D detections with corresponding 3D point clusters. The system selects the most representative LiDAR point in each matched cluster, such as the centroid or closest point, and computes the real-world distance to the object using Euclidean metrics. By combining the rich spatial structure of LiDAR with the semantic understanding of deep learning-based vision, this fusion system achieves improved accuracy in object detection and reliable distance estimation, even under occlusion and varying lighting conditions. This work demonstrates a cost-effective, accurate, and easily reproducible methodology for object detection and localization in 3D space, making it suitable for academic exploration as well as practical deployment in intelligent transportation systems. Future work may include multi-frame fusion, real-time tracking, and temporal consistency for dynamic object motion prediction.

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1. Introduction

Accurate localization is a fundamental requirement for autonomous vehicles and intelligent robotic systems. The ability to determine a vehicle's position in the world enables safe navigation, path planning, and interaction with the surrounding environment. Traditionally, Global Positioning System (GPS) technology has been the cornerstone of localization in outdoor environments. GPS provides absolute position data, which is particularly useful for long-range navigation and map-based planning.

However, GPS has notable limitations. It suffers from reduced accuracy or complete signal loss in urban canyons, tunnels, dense forests, or indoor environments. Multi-path interference from tall structures and atmospheric conditions can further degrade its reliability. For autonomous systems operating in such environments, sole reliance on GPS is not sufficient for ensuring safety and continuity of localization.

To overcome these limitations, relative localization techniques have been developed, which estimate the vehicle's position with respect to nearby objects and the immediate surroundings. These methods utilize data from onboard sensors—such as LiDAR, cameras, and radar—to perceive the environment and infer position through environmental context.

This project adopts such a relative localization approach by combining LiDAR point cloud clustering with visual object detection. By matching spatially clustered 3D LiDAR data with objects detected in 2D camera images, the system can estimate distances and localize the ego vehicle relative to its surroundings, even in GPS-denied scenarios. This sensor fusion strategy enhances both robustness and situational awareness, contributing to safer and more reliable autonomous navigation.

1.1 Motivation

Intelligent autonomous systems are transforming industries such as transportation, logistics, and urban automation. Their success relies heavily on accurate perception of the environment. This perception is achieved by integrating data from various sensors, each with unique advantages and limitations.

Cameras provide high-resolution visual data, making them suitable for detecting and classifying objects like traffic signs, vehicles, and pedestrians. They capture color, texture, and context—important for understanding scenes. However, cameras lack depth-sensing capabilities and are highly sensitive to lighting and weather conditions. In low light, fog, or glare, their performance deteriorates significantly.

LiDAR (Light Detection and Ranging) sensors compensate for these weaknesses by providing accurate 3D information about object shapes and distances. They generate dense point clouds that represent the geometry of the surroundings, independent of lighting. This makes LiDAR ideal for tasks requiring spatial understanding, such as mapping and obstacle avoidance. However, LiDAR units are often costly and may face challenges in heavy rain or fog, which can scatter laser pulses and reduce accuracy.

Radar sensors are resilient in adverse weather and are excellent for detecting the velocity and position of objects. While reliable, radar suffers from low spatial resolution and limited object classification abilities. Ultrasonic sensors are effective at short distances and are used for tasks like parking assistance but are limited in range and detail. Thermal cameras detect infrared radiation, making them useful for night-time or smoky environments, though they lack visual detail and can be expensive.

Because no single sensor is perfect, sensor fusion is a key strategy. This project fuses information from RGB cameras and LiDAR. Object detection is performed on 2D images using vision models like YOLO to identify cars, pedestrians, and other entities. In parallel, LiDAR data is clustered to find physical objects in 3D space. By applying calibration matrices, the 3D LiDAR points are projected onto the image plane, allowing a match between 2D detections and 3D clusters.

This fusion allows accurate distance measurement and relative localization of the ego vehicle with respect to its surroundings, independent of GPS. In environments like tunnels or dense urban areas where GPS fails, this method enables the vehicle to localize itself by understanding its spatial relationship with nearby objects.

By addressing the weaknesses of individual sensors through fusion, this project contributes to the development of autonomous systems that are more reliable and effective in real-world conditions.

1.2 Objectives

Perform 3D clustering of LiDAR points to group spatially close points representing individual objects, and conduct sensor fusion by matching these LiDAR clusters with image detections.

Detect and classify objects in camera images and measure the distance of detected objects from the ego vehicle.

Limit the average distance error between matched objects and ground truth positions, ensuring reliable localization accuracy.

1.3 Literature Survey

[1] The methodology of this work involves a decision-level fusion technique, which combines 3D point cloud data from LiDAR with 2D image data from a camera. First, LiDAR data are processed using the PointPillars deep learning network to generate 3D detection boxes, while the camera images are analyzed using the YOLOv5 network to identify 2D object bounding boxes. These detection results are then aligned using a joint extrinsic calibration process that projects the 3D LiDAR detections onto the 2D image plane. To associate the corresponding boxes, a center-point distance probability is calculated and followed by an Intersection-over-Union (IoU) matching strategy. The fusion of class confidence is further refined using an improved Dempster–Shafer (D–S) theory. An improved DeepSORT algorithm is applied, incorporating an unscented Kalman filter for non-linear motion prediction and data association improvements.

[2] Methodologies typically involve low-level fusion where raw data are combined, or high-level fusion where features or decisions from both sensors are merged. An example of this is 3D object detection using methods like the PointFusion network, where camera images and LiDAR point clouds are processed separately and fused at a later stage to estimate bounding boxes around objects. Additionally, for tasks such as occupancy grid mapping and moving object detection, the fusion process processes camera data for semantic cues while LiDAR contributes geometric structure, resulting in accurate environmental modeling even in dynamic scenarios.

[3] In autonomous vehicles, LiDAR-camera fusion enhances perception by combining LiDAR’s precise depth data with the camera’s rich visual information. The study performs intrinsic calibration using a checkerboard and extrinsic

calibration with a 3D planar marker. A projective transformation maps 3D LiDAR points onto the 2D camera plane, enabling low-level spatial alignment. This fusion allows accurate distance estimation by identifying LiDAR points within camera-detected object bounding boxes. Validated through simulations and real-road tests, the method improves the accuracy and reliability of object detection and depth estimation, demonstrating its effectiveness for real-time applications in autonomous driving under diverse conditions.

[4] This methodology estimates 3D pose of a UAV by fusing IMU, camera, and 2D/3D LiDAR data using an Extended Kalman Filter (EKF). The EKF predicts motion using high-rate IMU data and corrects drift with camera and LiDAR inputs. The camera provides visual motion cues but lacks depth accuracy, while 2D LiDAR offers precise planar position and yaw. 3D LiDAR enhances full spatial localization with rich depth data. Fusion addresses individual sensor limitations and improves accuracy, especially in GPS-denied settings. The approach shows that combining diverse sensor strengths through EKF significantly boosts robust 3D localization.

[5] Nguyen et al. (2025) propose a multi-layer LiDAR-camera fusion method to enhance object detection and localization in autonomous vehicles. Using the CARLA simulator, the system performs pixel-level fusion by projecting LiDAR points onto the camera image using calibrated parameters, enriching each pixel with depth information. Feature-level fusion then applies YOLOv8 to detect objects, associating bounding boxes with LiDAR points to compute precise 3D positions. Temporal and spatial synchronization is ensured through GPS signals and transformation matrices. The system adapts to varying environments by adjusting fusion weights and achieves ~ 0.54 m error and 30 FPS performance even in challenging conditions.

[6] Sensor fusion is crucial for robust perception in autonomous driving, with LiDAR-camera fusion offering a powerful combination of complementary strengths. LiDAR provides precise 3D spatial data for accurate localization, while cameras supply rich semantic information like color and texture. Fusion methods include early fusion—aligning LiDAR points with image pixels using calibration matrices—and late fusion, which merges results from independently processed sensor data. Deep learning models enhance this integration, enabling improved object detection and scene understanding. Literature shows that LiDAR-camera fusion significantly boosts perception accuracy and reliability, especially under poor lighting or occlusion, making it essential for safe autonomous navigation.

1.4 Problem Statement

This project addresses the limitations of single-sensor perception by fusing LiDAR and camera data. It focuses on clustering 3D LiDAR points and matching them with 2D camera image detections to improve object classification and estimate distances more accurately, ultimately enhancing object localization in autonomous driving and driver assistance systems.

1.5 Application in Societal Context

Accurate localization is essential for the safe and efficient operation of autonomous vehicles. In many environments, especially urban canyons, tunnels, or densely built areas, GPS signals can be weak or unreliable. In such cases, fusing data from multiple sensors such as LiDAR and cameras allows vehicles to determine their position with high precision. This enhanced situational awareness enables safer navigation, particularly in areas where traditional GPS-based methods may fail.

Improved localization also directly contributes to road safety. By accurately identifying the position of surrounding objects and their distances, autonomous systems can make better decisions, reducing the chances of collisions or unsafe maneuvers. Reliable tracking of dynamic objects ensures smoother navigation and better response to changing traffic conditions.

As cities evolve into smarter, more connected environments, autonomous vehicles with robust localization systems can play a key role in optimizing urban mobility. This includes coordinated traffic flows, adaptive traffic signaling, and integration with public transport networks, all of which lead to reduced congestion and better resource management.

Furthermore, enhanced localization expands the reach of autonomous vehicles to serve a wider demographic. People who are elderly, disabled, or without access to private transport can benefit from safe and reliable mobility solutions, promoting social inclusion and independence.

Lastly, improved positioning allows vehicles to minimize unnecessary detours and idling, reducing fuel consumption and emissions. This efficiency supports a more sustainable transportation system, aligning with global goals for reducing the environmental footprint of urban mobility.

2. System design

This section includes a functional block diagram that outlines the overall architecture, a detailed analysis of the KITTI dataset used for evaluation, and a comprehensive list of system requirements, providing the foundational setup necessary to implement the project.

2.1 Dataset

The KITTI dataset is a widely recognized benchmark suite used in autonomous driving and computer vision research. It was collected in real-world urban and highway environments using a vehicle equipped with multiple synchronized sensors. These include high-resolution stereo cameras, a 3D Velodyne LiDAR scanner, and GPS/IMU units for precise localization.

The KITTI dataset offers:

- Stereo RGB and monochrome camera images
- 3D LiDAR point clouds (.bin format)
- Calibration files for aligning different sensors
- GPS/IMU data for ground truth trajectory

The data is captured under varied lighting and traffic conditions, making it suitable for testing algorithms in realistic scenarios. The raw sequences contain around 6 hours of traffic data, covering over 390,000 stereo frames and 200 km of driving. KITTI is organized into multiple benchmark categories:

- Raw Data: Full sensor recordings with no labels; ideal for training and experimentation
- Object Detection: Annotated 2D and 3D bounding boxes for cars, pedestrians, and cyclists
- Tracking: Continuation of detection benchmark with temporal consistency across frames
- Stereo and Optical Flow: Benchmarks for depth estimation and motion tracking
- Road and Lane Detection: Semantic segmentation of drivable space
- Visual Odometry: Camera trajectory estimation with GPS/IMU ground truth

In this project, the Object Detection category is used. They allow the fusion of YOLO-based 2D image detections with 3D point cloud clustering to localize

surrounding objects accurately. The calibration files are crucial in projecting LiDAR points onto image planes, enabling sensor fusion.

2.2 Flow chart

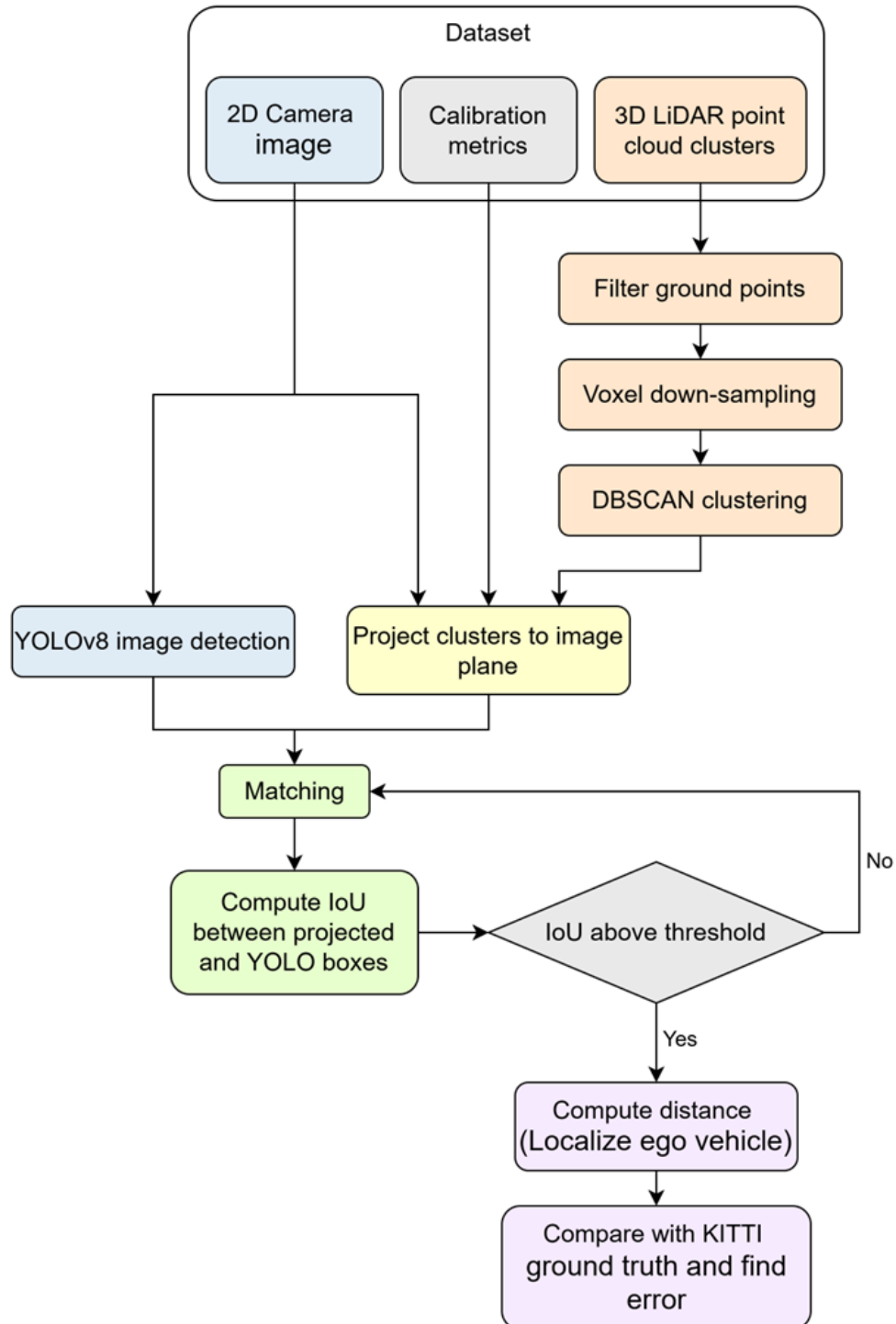


Fig. 1: Localization of an autonomous vehicle relative to surrounding objects

3. Implementation details

This section describes the methodology to develop the YOLO and LiDAR fusion-based object distance estimation system, including data preprocessing, sensor calibration, object detection, point cloud processing, clustering, and fusion techniques.

3.1 Methodology

The approach followed for implementing the object detection and distance estimation system using YOLOv8 and LiDAR data fusion is given below, detailing each stage from input handling to final output generation.

3.1.1 Data Loading and Preprocessing

The first step involves accessing and organizing the input data. For each frame in the KITTI dataset, the corresponding RGB image, LiDAR point cloud, calibration matrices, and ground truth label files are loaded. The RGB image is processed using the OpenCV library, whereas the LiDAR data, which is stored in binary format (.bin), is read and reshaped into an array of 3D coordinates (x, y, z). Calibration files containing intrinsic and extrinsic camera parameters are parsed to facilitate accurate projection between sensor modalities. Parsing of KITTI label files provides access to annotated object locations for validation.

3.1.2 LiDAR Point Cloud Processing and Clustering

Raw LiDAR data often includes irrelevant points (e.g., ground or background clutter), so initial filtering is performed to remove points that fall outside a defined range. To manage the density of the point cloud and reduce computational cost, voxel-based downsampling is applied. Once the point cloud is cleaned and simplified, the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm is used to group points into clusters. Each cluster ideally represents a physical object or part of one, such as a car, pedestrian, or cyclist. The geometric center and spread of these clusters are computed for use in projection and matching stages.

3.1.3 Object Detection Using Camera Images

The RGB image is passed through the YOLOv8 object detection model, which returns a list of bounding boxes, each with a corresponding class label (such as "car", "person", etc.) and a confidence score. This model is pre-trained on large datasets and is capable of detecting common road objects with high accuracy. The output of this step forms the 2D detection base, which will later be matched with 3D data for localization.

3.1.4 Projection of LiDAR Clusters onto Image Plane

To align the 3D LiDAR data with the 2D image space, a transformation process is carried out using the parsed calibration matrices. Each point in the clustered LiDAR data is projected into the camera frame, converting its 3D coordinates into corresponding pixel locations in the image. This process uses the rectification matrix, projection matrix, and transformation matrix to ensure precise alignment. Once projected, a 2D bounding box is drawn around the projected points for each cluster, allowing them to be directly compared with YOLO detections.

3.1.5 Matching Projected Clusters with Image Detections

Once both detection sets (YOLO boxes and projected clusters) are available in 2D space, the next step is to establish correspondence between them. This is achieved using the Intersection over Union (IoU) metric, which measures the overlap between two bounding boxes. For every projected LiDAR cluster, IoU is calculated with all YOLO boxes. The YOLO detection that yields the highest IoU (above a defined threshold) is selected as the match for that cluster. This association links the 3D position data from the LiDAR to the classification result from the camera image.

3.1.6 Visualization of Final Results

In the final stage, the image is updated to visually present the results. The original camera frame is displayed with both YOLO bounding boxes and the corresponding projected LiDAR clusters. Points belonging to each cluster are plotted within the matched bounding box, and the estimated distance from the sensor to each object is computed and displayed. This offers a clear, intuitive way to understand object presence, identity, and distance from the ego-vehicle. In case

no match is found for a cluster, a message is printed, ensuring full transparency of the system's behavior.

4. Results and conclusions

This section provides details about the results of this project. The yellow boxes represent YOLO detected objects (cars, truck, etc.). The green boxes represent LiDAR detected objects.

4.1 Impact of IoU over matching accuracy

IoU is a metric used to evaluate the accuracy of an object detector by measuring the overlap between the predicted bounding box and the ground truth box. It is defined as:

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

An IoU value ranges from 0 to 1, where 1 indicates a perfect match between the predicted and ground truth boxes. Here, area of overlap refers to overlap between the projected LiDAR cluster bounding box (in the image plane) and the YOLO-detected 2D bounding box from the image. A high IoU score indicates strong spatial overlap, enabling accurate association of 2D detections with corresponding 3D point clusters.

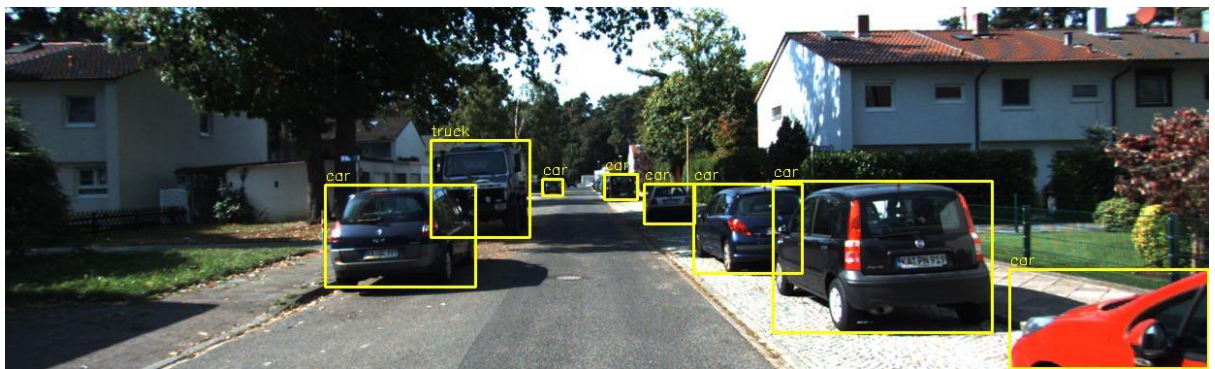


Fig. 2: YOLO detected objects (cars)

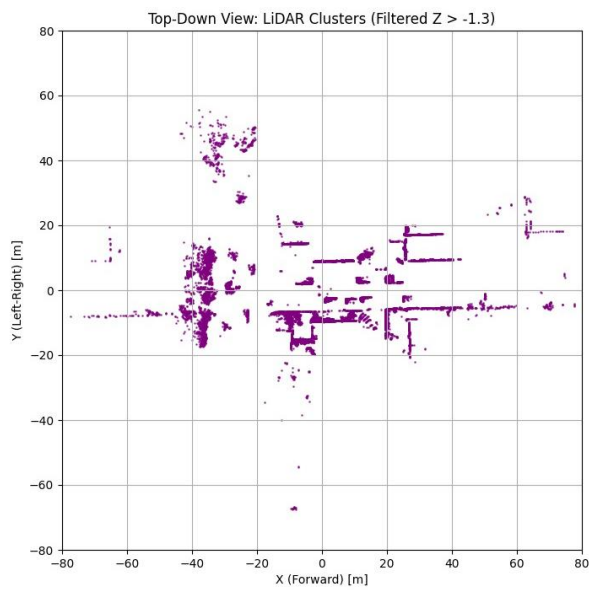


Fig. 3: LiDAR point cloud clusters

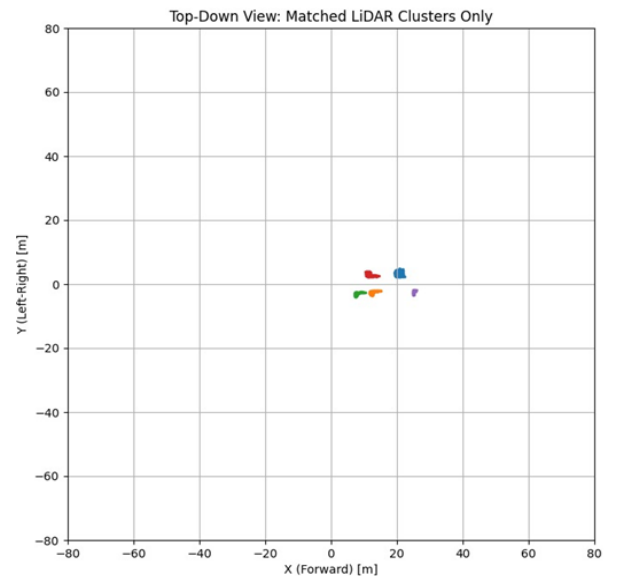


Fig. 4: LiDAR point cloud clusters matched with YOLO detected objects (IoU>0.5)

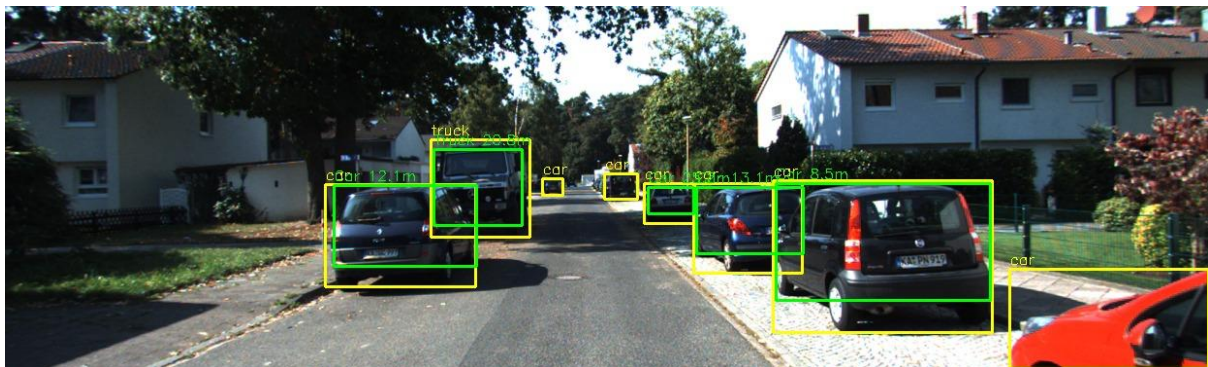


Fig. 5: LiDAR point cloud clusters matched with YOLO detected objects (IoU>0.5)

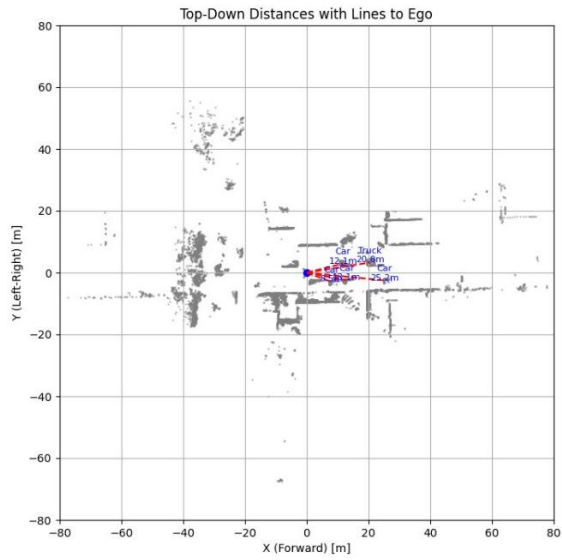


Fig. 6: Distance from ego vehicle to detected objects

IoU	Detected objects	Selected objects	Measured distance (in m)	Ground truth distance (in m)	Error (in m)
>0.3 and 0.5	5 cars 1 truck	Car 1	13.11	12.73	0.38
		Car 2	8.54	9.18	0.64
		Car 3	12.12	12.73	0.62
		Car 4	25.25	26.60	1.36
		Truck	20.80	22.59	1.79
>0.7	5 cars 1 truck	Car 1	13.11	12.73	0.38
		Car 2	8.54	9.18	0.64
		Car 3	12.12	12.73	0.62

Table 1: Distance from ego vehicle to detected objects and error

4.2 Performance Evaluation Across Multiple Images

Following are the images in which several objects are detected by camera and LiDAR but objects with $\text{IoU} > 0.5$ are considered for distance estimation. Table 2 presents the organized statistical results in a structured format for clearer interpretation.

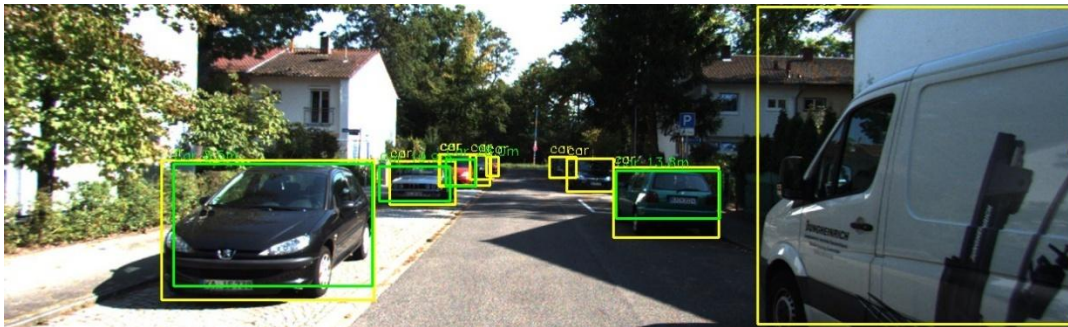


Image 1

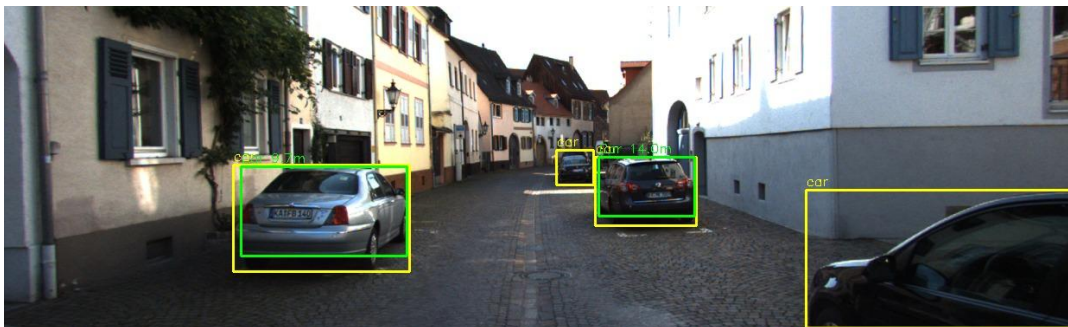


Image 2



Image 3



Image 4

Image	Detected objects	Selected objects	Measured distance (in m)	Ground truth distance (in m)	Error (in m)
1	9 cars 1 truck	Car 1	13.85	14.62	0.77
		Car 2	28.02	29.16	1.14
		Car 3	8.47	8.90	0.43
		Car 4	19.49	21.25	1.36
2	4 cars	Car 1	14.03	15.04	1.00
		Car 2	9.72	10.38	0.66
3	1 person 13 bicycles 1 potted plant	Person 1	8.91	8.73	0.18
4	1 person 1 car 1 bicycle 1 traffic light	1 car	22.40	22.93	0.53
		1 person	24.98	24.93	0.04

Table 2: Distance from ego vehicle to detected objects and error when IoU > 0.5

5. References

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